

AI Shopping Agent

Intelligent Conversational Ecommerce Platform

Project Type: AI Powered Ecommerce Assistant

Hackathon Submission

Frontend Deployment:

<https://ai-fashion-agent-ecru.vercel.app/>

Backend Deployment:

<https://ai-fashion-agent.onrender.com>

GitHub Repository:

<https://github.com/Deepti25sahu/ai-fashion-agent>

Deepti Kumari Sahu

Contents

1	Introduction	2
2	Problem Statement	2
3	Proposed Solution	2
4	Core Features	3
4.1	AI Fashion Search	3
4.2	Outfit Recommendation Engine	3
4.3	Recommendation Reasoning	3
4.4	Tradeoff Analysis	3
4.5	Voice Search	3
4.6	Dark Mode	3
4.7	Wishlist System	3
4.8	Product Comparison	3
5	User Flow	4
6	System Architecture	4
6.1	Frontend	4
6.2	Backend	4
6.3	Deployment	4
7	AI Agent Architecture	4
7.1	Intent Agent	4
7.2	Search Agent	5
7.3	Recommendation Agent	5
7.4	Ranking Agent	5
7.5	Outfit Agent	5
8	Technology Stack	5
9	Competitive Advantages	5
10	Future Enhancements	6
11	Conclusion	6

1 Introduction

AI Fashion Shopping Agent is an intelligent conversational ecommerce platform that transforms traditional online shopping into a modern AI-powered experience.

The system allows users to search fashion products using natural language instead of manually browsing through complex filters and categories.

Users can search using queries such as:

- Black dress for farewell
- Oversized streetwear hoodie
- Pink heels for party
- Casual airport outfit

The platform intelligently recommends products, suggests outfits, provides reasoning for recommendations, and improves the shopping journey through conversational AI.

2 Problem Statement

Traditional ecommerce platforms suffer from several usability problems:

- Users spend excessive time searching products manually.
- Product discovery is difficult for users unfamiliar with fashion terminology.
- Existing platforms provide poor personalization.
- Users struggle to compare products effectively.
- Fashion styling assistance is mostly unavailable.
- Too many options create decision fatigue.

The objective of this project is to solve these problems using conversational AI and intelligent recommendation systems.

3 Proposed Solution

The AI Fashion Shopping Agent provides:

- Natural language fashion search
- AI-powered outfit recommendations
- Smart product ranking
- Product comparison system
- Recommendation reasoning

- Voice-based fashion search
- Wishlist management
- Responsive premium ecommerce UI
- Follow-up conversational questions

The platform behaves like a personal AI stylist and shopping assistant.

4 Core Features

4.1 AI Fashion Search

Users can search using conversational queries instead of keyword-only search.

4.2 Outfit Recommendation Engine

The system suggests complementary outfit items dynamically.

4.3 Recommendation Reasoning

Products include AI-generated reasoning such as:

“Recommended because it matches elegant farewell fashion aesthetics.”

4.4 Tradeoff Analysis

The platform explains tradeoffs between products:

- Premium quality vs affordability
- Budget vs luxury
- Minimal vs trendy

4.5 Voice Search

Users can search products using speech recognition.

4.6 Dark Mode

Luxury dark/light theme support improves user experience.

4.7 Wishlist System

Users can save products locally using browser localStorage.

4.8 Product Comparison

Products can be compared side-by-side for smarter decisions.

5 User Flow

1. User enters a fashion query
2. AI analyzes intent
3. Product search begins
4. AI ranks products
5. Outfit suggestions are generated
6. User compares products
7. User saves wishlist
8. User clicks purchase links

6 System Architecture

6.1 Frontend

- Next.js 14
- TypeScript
- Tailwind CSS
- Framer Motion
- Lucide Icons

6.2 Backend

- FastAPI
- Python
- AI Agents
- Recommendation Engine

6.3 Deployment

- Vercel (Frontend)
- Render (Backend)

7 AI Agent Architecture

7.1 Intent Agent

Understands user search intent.

7.2 Search Agent

Generates optimized fashion search queries.

7.3 Recommendation Agent

Filters relevant products.

7.4 Ranking Agent

Ranks products intelligently based on relevance and quality.

7.5 Outfit Agent

Suggests complementary fashion products.

8 Technology Stack

Technology	Purpose
Next.js 14	Frontend Framework
TypeScript	Type Safety
Tailwind CSS	UI Styling
Framer Motion	Animations
FastAPI	Backend API
Python	AI Logic
SerpAPI	Product Search
Vercel	Frontend Deployment
Render	Backend Deployment

9 Competitive Advantages

Compared to traditional ecommerce platforms, this system provides:

- Conversational shopping experience
- AI-powered fashion guidance
- Product reasoning
- Smart recommendations
- Modern premium UI
- Intelligent product comparison

10 Future Enhancements

Future upgrades may include:

- AI Virtual Try-On
- Fashion Trend Prediction
- AI Stylist Chatbot
- Personalized User Profiles
- Checkout and Payment Gateway
- Image-Based Fashion Search
- Firebase Authentication
- Fashion Reel Recommendations
- Recommendation Learning System

11 Conclusion

AI Fashion Shopping Agent demonstrates how conversational AI can transform ecommerce into a more intelligent, personalized, and engaging experience.

The project combines AI recommendation systems, premium frontend engineering, conversational interaction, and modern ecommerce design to create a startup-level fashion shopping assistant.

The platform successfully bridges the gap between traditional ecommerce browsing and intelligent AI-driven product discovery.